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FORM TP 2010005

JANUARY 2010

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

CHEMISTRY

Paper 02 – General Proficiency

2 hours and 30 minutes

READ THE FOLLOWING DIRECTIONS CAREFULLY.

- 1. This paper consists of SEVEN questions in THREE sections.**
- 2. Answer the THREE compulsory questions in Section A. Write your answers in this answer booklet.**
- 3. Answer the TWO compulsory questions in Section B. Write your answers in the spaces provided in this answer booklet.**
- 4. Answer ONE question from Section C. Write your answers in the space provided in this answer booklet.**
- 5. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.**
- 6. The use of silent, non-programmable calculators is allowed.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer ALL questions in this section.

Write your answers in the spaces provided in this booklet.

Do NOT spend more than 30 minutes on Question 1.

1. (a) Copper(II) oxide can react with dilute sulphuric acid to produce either the anhydrous or the hydrated form of a salt as one of the products. The type of salt produced (anhydrous or hydrated) depends on the procedure used.

Table 1 shows 5 experiments with the different masses of an anhydrous copper(II) salt that can be formed by reacting different masses of copper(II) oxide with 50 cm³ of 1 mol dm⁻³ sulphuric acid. One reading for Experiment 3 is missing from Table 1.

TABLE 1: MASSES OF ANHYDROUS COPPER(II) SALT FORMED FROM COPPER(II) OXIDE USING 1 MOL DM⁻³ H₂ SO₄

Experiment Number	1	2	3	4	5
Mass of copper(II) oxide used (g)	1.58	2.36	3.58	4.00	4.50
Mass of anhydrous salt produced (g)	3.16	4.72		8.00	8.00

- (i) Explain how a salt is formed.

(2 marks)

- (ii) What is the difference between 'anhydrous salts' and 'hydrated salts'?

(2 marks)

- (iii) Write a **balanced** chemical equation to show the production of the **anhydrous** salt from the reaction between copper(II) oxide and dilute sulphuric acid.

(2 marks)

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- (iv) Given that the relative formula mass of copper(II) oxide is 80, complete the data in Table 1 by calculating the mass of the salt formed in Experiment 3. (Relative Atomic Mass: Cu = 64; S = 32; O = 16; H = 1)

Show your working.

(4 marks)

- (v) Using the data from Table 1 and the axes provided on page 4, plot a graph of the mass of anhydrous salt produced against the mass of copper(II) oxide used.

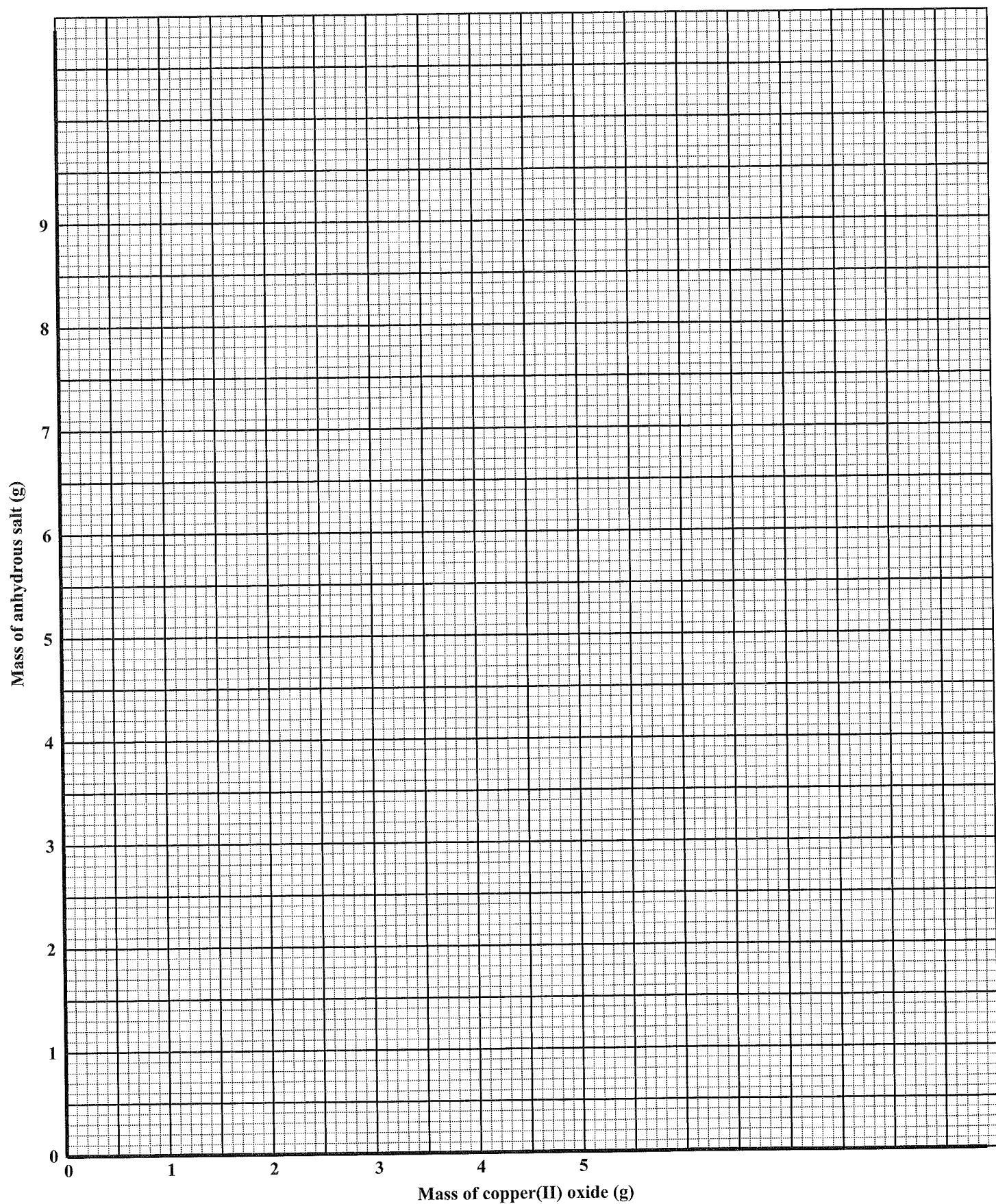
(3 marks)

- (vi) Using the graph plotted in (v) above, determine the mass of salt that would be formed from 2.95 g of copper(II) oxide.

(1 mark)

- (vii) Explain why there is no change in the mass of salt produced for Experiments 4 and 5.

(1 mark)



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- (b) A student conducts a number of tests on Substance X. Some of the observations and inferences are recorded in Table 2. Complete Table 2 to show the remaining observations and inferences made by the student.

TABLE 2: TESTS ON SUBSTANCE X

Test	Observation	Inference
A small sample of Solid X is gently heated in a dry test tube. Damp red litmus is used to test for any gas.		Alkaline gas is produced.
	(1 mark)	
Approximately 2 cm ³ of aqueous sodium hydroxide is added to a sample of Solid X and warmed.	A gas evolves which forms dense white fumes with hydrogen chloride.	
		(Equation required) (2 marks)
A small amount of Solid X is dissolved in about 5 cm ³ of water. To this solution, sodium hydroxide is added gradually until in excess.		X contains iron(II) ions.
	(2 marks)	(Ionic equation required) (2 marks)

- (c) Describe an appropriate method to produce a pure solid sample of silver chloride starting with silver nitrate solution. (Hint! All silver salts are decomposed by light.)

(3 marks)

Total 25 marks

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2. V and Q are two different forms of the element carbon. R is an ionic solid. Table 3 presents some properties of V, Q and R. Use this information to answer the questions that follow.

TABLE 3: INCOMPLETE TABLE SHOWING PROPERTIES OF V, Q AND R

Property	V	Q	R
Appearance	Transparent, colourless, solid	Black, opaque, solid, shiny	White, solid
Melting Point (°C)	3823 °C	3925 °C	801 °C
Electrical Conductivity	Electrical non-conductor		Electrical conductor when molten but non-conductor in solid state
Structure		Giant molecular or giant covalent	Giant ionic

- (a) (i) Complete Table 3 by writing in the table, the
- structure of V
 - electrical conductivity of Q. (2 marks)
- (ii) State the term used to describe different forms of an element such as the two forms of carbon, V and Q.

_____ (1 mark)

- (b) (i) Identify the forms of carbon, V and Q.

V _____

Q _____

(2 marks)

- (ii) Figure 1 represents a partially drawn structure of Q. Complete the figure to show the bonding within and between the layers.

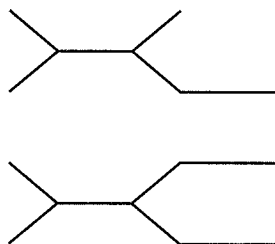


Figure 1. Partially drawn structure of Q

(2 marks)

- (iii) State ONE use of Element V and ONE use of Element Q and indicate the property upon which EACH use is based.

Use of V

Property of V

Use of Q

Property of Q

(4 marks)

- (iv) Explain why R will conduct electricity when molten but NOT in the solid state.

(2 marks)

- (c) Iodine and chlorine are halogens. They combine to form a gaseous compound, iodine monochloride (ICl).

Using dot cross diagrams, show the bonding in iodine monochloride.

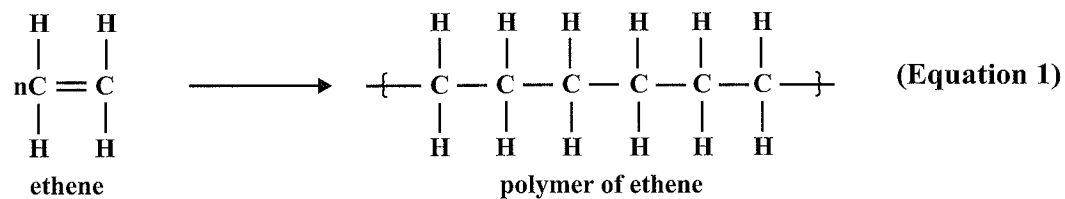
(2 marks)

Total 15 marks

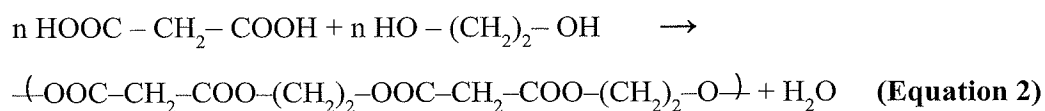
3. (a) Define the term 'polymer'.

(2 marks)

- (b) The polymerisation of ethene is represented in **Equation 1** below.



The polymerisation reaction between a di-acid, $\text{HOOC}-\text{CH}_2-\text{COOH}$ and a di-alcohol, $\text{HO}-(\text{CH}_2)_2-\text{OH}$ is represented in the unbalanced **Equation 2** below.



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- (i) Which of the Equations in (b) above shows addition polymerisation?

(1 mark)

- (ii) Give the name of EACH of the polymers shown in Equation 1 and Equation 2.

Polymer in Equation 1

Polymer in Equation 2

(2 marks)

- (iii) State ONE use of EACH of the polymers named in (ii) above.

Polymer in Equation 1

Polymer in Equation 2

(2 marks)

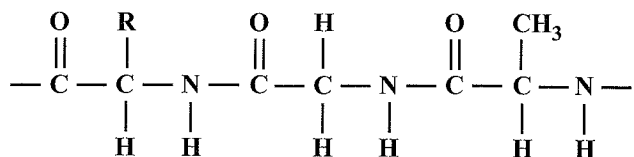
- (c) (i) Ethanoic acid, CH_3COOH , is a weak, monobasic, organic acid. Write a balanced equation for the reaction of ethanoic acid with magnesium metal, Mg (s).

(2 marks)

- (ii) Would this reaction take place at room temperature?

(1 mark)

- (d) During digestion, proteins undergo hydrolysis to form amino acids, which are absorbed by the body. A partial structure of a protein molecule, where R is an alkyl group is shown below.



- (i) On the diagram, circle the peptide bond (link) that is found in proteins.
(1 mark)
- (ii) Based on the partial protein structure given above, draw the fully displayed structure of an amino acid that would be produced upon hydrolysis.
(2 marks)
- (iii) State TWO conditions under which proteins can be hydrolysed.

(2 marks)

Total 15 marks

SECTION B

Answer ALL questions in this section.

Write your answers in the space provided after EACH question in this answer booklet.

4. (a) Figure 2 shows the apparatus used to investigate the conducting properties of some substances, when placed in contact with electrodes.

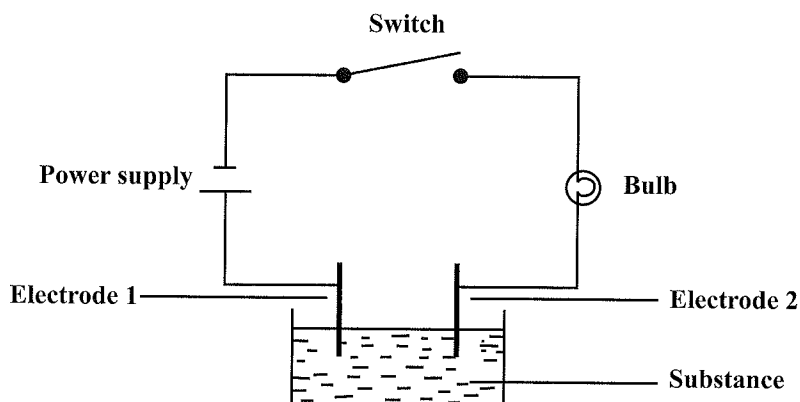


Figure 2. Apparatus for investigating the conducting properties of substances

The investigation is carried out on two pairs of substances as indicated below.

- (i) Pair 1: solid lead bromide
molten lead bromide

State whether or not the bulb will glow for EACH substance. Explain your answer for EACH substance. **(4 marks)**

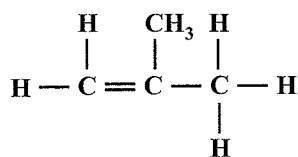
- (ii) Pair 2: 1 mol dm^{-3} hydrochloric acid
 1 mol dm^{-3} ethanoic acid

The bulb glows for both substances, but glows more brightly for one of the substances.

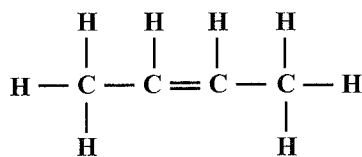
- a) State which substance causes the bulb to glow more brightly.
- b) Explain why the substance you have stated at a) causes the bulb to glow brighter than the other substance. **(3 marks)**
- (iii) Write a balanced ionic equation for the reaction occurring at the cathode for any of the substances in Pair 2. **(2 marks)**

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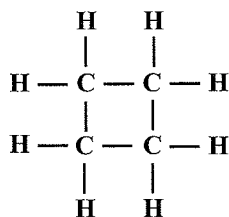
- (b) The structures of four hydrocarbons, **a**, **b**, **c** and **d** are shown below.



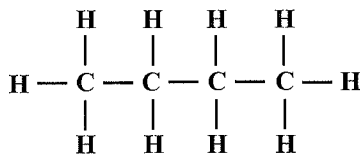
a



b



c



d

- (i) Give the name for structure **a**. (1 mark)
- (ii) State which of these hydrocarbons are isomers. (2 marks)
- (iii) Which TWO hydrocarbons belong to the same homologous series? Give ONE reason for your answer. (2 marks)
- (iv) Write the general formula of the isomers identified in (iii) above. (1 mark)

Total 15 marks

Write your answer to Question 4 here.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

5. (a) Hydrogen, chlorine and nitrogen are all diatomic gases. The following chemical test(s) can be used to distinguish among them.

Test 1	A lighted splint
Test 2	A piece of moist litmus

Explain how the tests above can be used to distinguish among the gases, hydrogen, chlorine and nitrogen. Include observations in your answer. **(5 marks)**

- (b) Chlorine is industrially manufactured by the electrolysis of brine.
- (i) Explain what occurs at the anode and the cathode in the electrolysis of brine. **(4 marks)**
 - (ii) Write an ionic equation for the reaction which occurs at the anode. **(2 marks)**
 - (iii) Suggest why brine is used instead of dilute sodium chloride. Give an equation to justify your answer. **(4 marks)**

Total 15 marks

SECTION C

Answer EITHER Question 6 OR Question 7.

Write your answer in the space provided at the end of this booklet.

- 6.**
- (a)
 - (i) Describe the chemical composition of clay. **(1 mark)**
 - (ii) When clay is fired, a rigid glass-like framework is formed. Suggest a possible source of 'glass' in fired clay. **(1 mark)**
 - (iii) Clay is used extensively in pottery to make items of different shapes. Relate this use of clay to ONE of its properties. **(2 marks)**
 - (b) A mason advised one of his clients to keep the freshly completed concrete floor moist while hardening, to prevent cracks from forming. The client was concerned that this practice would stop the concrete from setting. From your knowledge of the composition of cement and its use in making concrete, explain the reason for the mason's advice. **(3 marks)**
 - (c) Fibres are abundant in nature and are produced by both plants and animals.
 - (i) Name ONE plant fibre and ONE animal fibre used in making fabric, **and** state the chemical composition of plant fibres and animal fibres. **(4 marks)**
 - (ii) Suggest TWO chemical tests which may be used to distinguish an animal fibre from a plant fibre. Include the expected observations. **(4 marks)**

Total 15 marks

7. There are many substances that are essential for healthy plant growth.

- (a) Name TWO elements that are essential for plant growth. For EACH of the elements named, identify ONE effect of its deficiency. (4 marks)
- (b) Figure 3 represents part of the nitrogen cycle. Some of the names of the reactions and compounds formed have been replaced by the letters A, B, C and D.

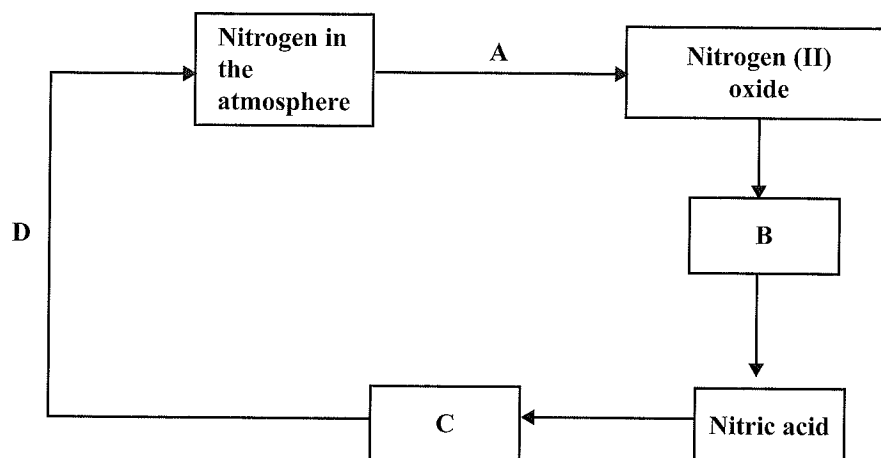


Figure 3. Incomplete Nitrogen Cycle

- (i) Write the names of the type of chemical reactions taking place at A and D. (2 marks)
- (ii) Write the name of the formula of compound B. (2 marks)
- (iii) Name the type of compound labelled C. (1 mark)
- (c) Lime is added to soil to increase the pH and reduce problems caused by acid soils, but it can also cause nitrogen to be lost from the soil. Suggest ONE way in which lime can cause nitrogen to be lost from the soil. Include a balanced ionic equation in your answer. (3 marks)
- (d) Hydroponics is an alternative method of growing crops.
- (i) State TWO advantages of using hydroponics. (2 marks)
- (ii) Suggest ONE possible limitation of using hydroponics. (1 mark)

Total 15 marks